

ANGEL FIRE, NM, USA

WMO: 720411

Lat: 36.422N

Lon: 105.290W

Elev: 2554

StdP: 74.18

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 00137

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -25.8	(c) -21.7	(d) -28.5	(e) 0.4	(f) -22.3	(g) -24.6	(h) 0.6	(i) -17.2	(j) 15.3	(k) -0.2	(l) 13.6	(m) 0.3	(n) 0.2	(o) 160	(p) 0.717

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.6	26.3	11.4	25.1	11.3	23.9	11.0	14.4	20.9	13.9	20.3	13.2	19.8	5.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	12.2	16.3	11.6	11.7	15.8	11.0	11.2	15.5	49.8	21.0	47.8	20.4	45.9	19.6	16.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.7	(b) 11.1	(c) 9.4	(e) -33.4	(f) 28.5	(g) 2.8	(h) 1.1	(i) -35.4	(j) 29.2	(k) -37.1	(l) 29.9	(m) -38.7	(n) 30.5	(o) -40.7	(p) 31.3		
(4) 12.7	(5) 11.1	(6) 9.4	(7) -33.6	(8) 15.7	(9) 2.8	(10) 0.5	(11) -35.6	(12) 16.1	(13) -37.2	(14) 16.4	(15) -38.7	(16) 16.7	(17) -40.8	(18) 17.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	4.4	-6.5	-4.1	0.6	3.9	7.5	12.6	15.0	13.9	10.9	5.1	-0.7	-5.3
	DBStd	8.29	5.19	5.29	4.52	3.53	3.17	2.28	1.49	1.70	2.35	3.75	4.54	5.40
	HDD10.0	2443	510	394	291	184	89	4	0	0	16	156	322	476
	HDD18.3	5068	768	627	549	434	337	171	105	138	223	409	572	734
	CDD10.0	416	0	0	0	1	11	83	153	121	43	4	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	371	0	0	0	0	0	146	133	70	22	0	0	0
	CDH26.7	21	0	0	0	0	0	14	6	1	0	0	0	0
Wind	WSAvg	3.2	3.1	4.1	4.0	4.4	3.7	3.1	2.1	2.1	2.3	3.1	3.0	3.6
Precipitation	PrecAvg	491	24	29	35	38	39	36	81	77	44	36	25	27
	PrecMax	648	57	48	77	109	107	82	174	152	83	92	51	63
	PrecMin	364	4	5	8	4	1	0	33	28	8	2	2	4
	PrecStd	75	14	14	18	29	31	21	36	29	22	21	14	17
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	11.4	17.1	18.1	22.8	28.3	27.4	27.0	25.8	21.1	15.9	10.4
		MCWB	1.2	2.4	4.9	5.8	8.0	10.9	12.0	11.4	10.6	7.8	4.5	1.3
	2%	DB	6.6	9.5	14.8	17.0	21.3	26.7	26.1	25.1	23.7	18.9	13.6	7.9
		MCWB	0.0	1.7	4.1	5.3	7.4	10.6	12.2	11.7	10.3	7.0	3.6	0.6
	5%	DB	4.9	7.5	12.4	15.2	19.9	25.0	25.0	23.8	22.3	17.2	11.8	5.9
		MCWB	-0.7	0.9	2.8	4.6	7.1	9.9	12.2	11.8	9.9	6.1	3.0	-0.1
	10%	DB	2.8	5.2	10.1	13.7	17.8	23.6	23.3	22.5	20.6	15.7	9.1	3.7
		MCWB	-1.4	-0.5	1.8	4.0	6.6	9.6	12.4	11.9	9.9	5.9	1.8	-1.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.5	3.6	5.5	7.5	10.6	14.2	15.2	15.0	13.5	11.2	5.8	2.9
		MCDB	6.4	9.8	14.7	15.9	17.7	21.5	22.0	21.4	19.1	16.1	12.8	7.2
	2%	WB	1.2	2.2	4.4	6.4	9.3	13.1	14.6	14.2	12.4	9.2	4.7	1.5
		MCDB	5.0	8.0	13.3	14.5	16.8	21.0	21.2	20.4	18.9	15.2	11.5	5.9
	5%	WB	0.0	1.1	3.2	5.3	8.2	12.2	14.1	13.7	11.7	7.7	3.5	0.4
		MCDB	3.6	6.6	11.3	13.5	16.5	20.2	20.6	20.0	18.3	13.6	9.9	4.6
	10%	WB	-1.1	0.0	2.2	4.4	7.3	11.3	13.5	13.0	10.9	6.6	2.4	-0.6
		MCDB	2.6	5.0	9.2	12.7	16.1	19.6	20.1	19.5	17.3	13.6	8.6	3.4
Mean Daily Temperature Range	5% DB	MDBR	16.8	14.4	15.4	15.7	17.3	20.7	17.6	17.8	18.7	17.8	17.0	14.8
		MCDBR	19.6	16.6	19.6	19.7	21.6	24.0	20.8	20.7	22.3	21.9	21.1	17.3
		MCWBR	13.8	10.0	10.1	10.1	10.4	10.8	8.9	9.3	10.6	11.5	12.3	11.1
	5% WB	MCDBR	16.3	14.7	17.6	18.2	18.8	19.4	16.1	16.3	17.7	17.6	18.3	14.6
		MCWBR	11.9	9.0	9.3	9.6	9.8	9.7	7.5	7.9	9.1	10.1	10.9	9.8
Clear-Sky Solar Irradiance	taub	0.222	0.228	0.257	0.277	0.286	0.303	0.342	0.320	0.281	0.264	0.233	0.228	
	taud	2.667	2.662	2.569	2.524	2.555	2.494	2.459	2.490	2.611	2.621	2.656	2.641	
	Ebn at Noon	1015	1039	1022	1008	995	972	932	950	981	978	990	985	
	Edh at Noon	67	76	92	103	101	108	111	105	88	79	67	64	
All-Sky Solar Radiation	RadAvg	3.14	4.07	5.33	6.48	7.19	7.49	6.52	6.13	5.57	4.50	3.40	2.74	
	RadStd	0.19	0.29	0.40	0.37	0.50	0.41	0.32	0.31	0.31	0.30	0.20	0.14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (13 neighbors)	+0.30	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+62	+26

Nomenclature: See separate page